Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Pimentel Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
11391 Franklin BLVD	Elk Grove	Sacramento	95757

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 07/03/1953

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

X132-X251-X007-XXXX

B. OPERATORS

Pimentel, Eddie

Operator name: Pimentel, Eddie

Telephone no.: Landline: ; Cellular: (123) 000-0000

Mailing Address Number and Street	City	State	Zip Code
11375 Bruceville RD	Elk Grove	CA	95757

This operator is responsible for paying permit fees.

C. OWNERS

Nunes/E&L Land Group, Tony & Filemina

Legal owner name: Nunes/E&L Land Group, Tony & Filemina

Telephone no.: Landline: ; Cellular: (123) 000-0000

Mailing Address Number and Street	City	State	Zip Code
11375 Bruceville RD	Elk Grove	CA	95757

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	100	80	0	0
Number under roof	350	30	0	0	0	0
Maximum number	350	30	100	80	0	0
Average number	350	30	100	80	0	0
Avg live weight (lbs)	1,400	1,500	1,050	600		

Predominant milk cow breed: Holstein

Average milk production: 62 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 10,884.69 *tons per reporting period*

Total nitrogen from manure: 135,643.55 *lbs per reporting period*

After ammonia losses (30% loss applied): 94,950.49 *lbs per reporting period*

Total phosphorus from manure: 22,695.74 *lbs per reporting period*

Total potassium from manure: 65,188.63 *lbs per reporting period*

Total salt from manure: 171,696.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 19,152,000 *gallons*

Total nitrogen generated: 35,291.84 *lbs*

Total phosphorus generated: 4,516.27 *lbs*

Total potassium generated: 49,533.25 *lbs*

Total salt generated: 412,176.24 *lbs*

	19,152,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	19,152,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well #1	Ground water
Ag Well #2	Ground water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/09/2023	Liquid commercial fertilizer *4-10-10*	1,332.40 *gal*			4.000000	10.000000	10.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	444.76	1,111.89	1,111.89	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	444.76	1,111.89	1,111.89	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	18	18	2	both	X132-X251-X007-XXXX
2	23	23	2	both	X132-X251-X007-XXXX
3	31	31	2	both	X132-X251-X007-XXXX
4	80	80	0	none	X132-X251-X007-XXXX
5	37	37	2	both	X132-X251-X007-XXXX
6	31	31	0	none	X132-X251-X007-XXXX
Totals for areas that were used for application	109	109	8		
Totals for areas that were not used for application	111	111	0		
Land application area totals	220	220	8		

B. CROPS AND HARVESTS

1

Field name: 1

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	270.00 *ton*	Dry-weight		71.3	27,700.00	1,780.00	17,700.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.00	238.50	15.33	152.40	1,188.18

1

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/30/2023	540.00 *ton*	Dry-weight		76.0	22,400.00	1,330.00	14,700.00		8.45

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	322.56	19.15	211.68	1,216.80

2

Field name: 2

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 23 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	345.00 *ton*	Dry-weight		63.6	13,100.00	1,900.00	11,800.00		9.94

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.00	143.05	20.75	128.86	1,085.45

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/30/2023	690.00 *ton*	Dry-weight		76.8	23,500.00	1,540.00	10,900.00		5.31

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	327.12	21.44	151.73	739.15

3

Field name: 3

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	465.00 *ton*	Dry-weight		65.1	20,400.00	1,620.00	10,300.00		9.48

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.00	213.59	16.96	107.84	992.56

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 31 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	930.00 *ton*	Dry-weight		73.0	16,100.00	1,440.00	8,600.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	260.82	23.33	139.32	1,393.20

5

Field name: 5

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	555.00 *ton*	Dry-weight		75.0	21,800.00	1,880.00	13,600.00		12.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.00	163.50	14.10	102.00	967.50

5

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	1,110.00 *ton*	Dry-weight		70.7	15,000.00	2,280.00	13,400.00		6.18

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	263.70	40.08	235.57	1,086.44

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/10/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	131.68	1.88	38.55	1,301.82	1,080,000.00 *gal*
Ag Well #1	Ground water	7.03	0.00	0.00	552.77	2,592,000.00 *gal*
Application event totals		138.71	1.88	38.55	1,854.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.79	1.26	25.70	867.88	720,000.00 *gal*
Ag Well #1	Ground water	4.69	0.00	0.00	368.52	1,728,000.00 *gal*
Application event totals		92.48	1.26	25.70	1,236.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.79	1.26	25.70	867.88	720,000.00 *gal*
Ag Well #1	Ground water	4.69	0.00	0.00	368.52	1,728,000.00 *gal*
Application event totals		92.48	1.26	25.70	1,236.40	

1 - 06/17/2023: Corn, silage

Field name: 1

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	63.12	155.22	974.34	7,023.10	180.00 *ton*
Application event totals		63.12	155.22	974.34	7,023.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Liquid commercial fertilizer	13.90	34.50	34.50	0.00	
Application event totals		13.90	34.50	34.50	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	14.06	0.00	0.00	1,105.55	5,184,000.00 *gal*
Application event totals		14.06	0.00	0.00	1,105.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	12.89	0.00	0.00	1,013.42	4,752,000.00 *gal*
Application event totals		12.89	0.00	0.00	1,013.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	11.72	0.00	0.00	921.29	4,320,000.00 *gal*
Application event totals		11.72	0.00	0.00	921.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.38	12.78	70.10	433.94	720,000.00 *gal*
Ag Well #1	Ground water	7.03	0.00	0.00	552.77	2,592,000.00 *gal*
Application event totals		41.41	12.78	70.10	986.71	

1 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.38	12.78	70.10	433.94	720,000.00 *gal*
Ag Well #1	Ground water	7.03	0.00	0.00	552.77	2,592,000.00 *gal*
Application event totals		41.41	12.78	70.10	986.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	11.72	0.00	0.00	921.29	4,320,000.00 *gal*
Application event totals		11.72	0.00	0.00	921.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.11	18.10	257.86	1,126.58	540,000.00 *gal*
Ag Well #1	Ground water	8.20	0.00	0.00	644.90	3,024,000.00 *gal*
Application event totals		83.31	18.10	257.86	1,771.48	

2 - 11/10/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.06	1.47	30.17	1,018.82	1,080,000.00 *gal*
Ag Well #1	Ground water	5.50	0.00	0.00	432.60	2,592,000.00 *gal*
Application event totals		108.56	1.47	30.17	1,451.42	

2 - 11/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.70	0.98	20.12	679.21	720,000.00 *gal*
Ag Well #1	Ground water	3.67	0.00	0.00	288.40	1,728,000.00 *gal*
Application event totals		72.37	0.98	20.12	967.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	3.67	0.00	0.00	288.40	1,728,000.00 *gal*
Application event totals		3.67	0.00	0.00	288.40	

2 - 06/17/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	63.12	155.22	974.34	7,023.10	230.00 *ton*
Application event totals		63.12	155.22	974.34	7,023.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Liquid commercial fertilizer	13.90	34.50	34.50	0.00	
Application event totals		13.90	34.50	34.50	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	14.67	0.00	0.00	1,153.61	6,912,000.00 *gal*
Application event totals		14.67	0.00	0.00	1,153.61	

2 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	13.75	0.00	0.00	1,081.51	6,480,000.00 *gal*
Application event totals		13.75	0.00	0.00	1,081.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	28.25	10.51	57.60	356.59	756,000.00 *gal*
Ag Well #1	Ground water	8.99	0.00	0.00	706.59	4,233,600.00 *gal*
Application event totals		37.24	10.51	57.60	1,063.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	13.75	0.00	0.00	1,081.51	6,480,000.00 *gal*
Application event totals		13.75	0.00	0.00	1,081.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.37	18.89	269.07	1,175.56	720,000.00 *gal*
Ag Well #1	Ground water	9.17	0.00	0.00	721.01	4,320,000.00 *gal*
Application event totals		87.54	18.89	269.07	1,896.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	12.23	0.00	0.00	961.34	5,760,000.00 *gal*
Application event totals		12.23	0.00	0.00	961.34	

2 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.37	18.89	269.07	1,175.56	720,000.00 *gal*
Ag Well #1	Ground water	9.17	0.00	0.00	721.01	4,320,000.00 *gal*
Application event totals		87.54	18.89	269.07	1,896.56	

3 - 11/12/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.46	1.09	22.39	755.90	1,080,000.00 *gal*
Ag Well #1	Ground water	4.08	0.00	0.00	320.96	2,592,000.00 *gal*
Application event totals		80.54	1.09	22.39	1,076.86	

04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	50.97	0.73	14.92	503.93	720,000.00 *gal*
Ag Well #1	Ground water	2.72	0.00	0.00	213.98	1,728,000.00 *gal*
Application event totals		53.70	0.73	14.92	717.91	

04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	50.97	0.73	14.92	503.93	720,000.00 *gal*
Ag Well #1	Ground water	2.72	0.00	0.00	213.98	1,728,000.00 *gal*
Application event totals		53.70	0.73	14.92	717.91	

3 - 06/19/2023: Corn, silage

3 - 06/19/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	63.12	155.22	974.34	7,023.10	310.00 *ton*
Application event totals		63.12	155.22	974.34	7,023.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Liquid commercial fertilizer	13.90	34.50	34.50	0.00	
Application event totals		13.90	34.50	34.50	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	10.88	0.00	0.00	855.91	6,912,000.00 *gal*
Application event totals		10.88	0.00	0.00	855.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	10.20	0.00	0.00	802.41	6,480,000.00 *gal*
Application event totals		10.20	0.00	0.00	802.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	27.95	10.39	56.98	352.75	1,008,000.00 *gal*
Ag Well #1	Ground water	5.71	0.00	0.00	449.35	3,628,800.00 *gal*
Application event totals		33.66	10.39	56.98	802.10	

3 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	27.95	10.39	56.98	352.75	1,008,000.00 *gal*
Ag Well #1	Ground water	5.71	0.00	0.00	449.35	3,628,800.00 *gal*
Application event totals		33.66	10.39	56.98	802.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	10.20	0.00	0.00	802.41	6,480,000.00 *gal*
Application event totals		10.20	0.00	0.00	802.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.40	19.62	279.49	1,221.06	1,008,000.00 *gal*
Ag Well #1	Ground water	5.71	0.00	0.00	449.35	3,628,800.00 *gal*
Application event totals		87.12	19.62	279.49	1,670.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	10.20	0.00	0.00	802.41	6,480,000.00 *gal*
Application event totals		10.20	0.00	0.00	802.41	

5 - 11/12/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

5 - 11/12/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	64.06	0.92	18.76	633.32	1,080,000.00 *gal*
Ag Well #1	Ground water	3.42	0.00	0.00	268.92	2,592,000.00 *gal*
Application event totals		67.48	0.92	18.76	902.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	42.71	0.61	12.50	422.21	720,000.00 *gal*
Ag Well #1	Ground water	2.28	0.00	0.00	179.28	1,728,000.00 *gal*
Application event totals		44.99	0.61	12.50	601.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	42.71	0.61	12.50	422.21	720,000.00 *gal*
Ag Well #1	Ground water	2.28	0.00	0.00	179.28	1,728,000.00 *gal*
Application event totals		44.99	0.61	12.50	601.49	

5 - 06/18/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 06/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	63.12	155.22	974.34	7,023.10	370.00 *ton*
Application event totals		63.12	155.22	974.34	7,023.10	

5 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Liquid commercial fertilizer	13.90	34.50	34.50	0.00	
Application event totals		13.90	34.50	34.50	0.00	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	10.26	0.00	0.00	806.75	7,776,000.00 *gal*
Application event totals		10.26	0.00	0.00	806.75	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	10.26	0.00	0.00	806.75	7,776,000.00 *gal*
Application event totals		10.26	0.00	0.00	806.75	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.07	7.46	40.92	253.33	864,000.00 *gal*
Ag Well #1	Ground water	6.38	0.00	0.00	501.98	4,838,400.00 *gal*
Application event totals		26.46	7.46	40.92	755.30	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.07	7.46	40.92	253.33	864,000.00 *gal*
Ag Well #1	Ground water	6.38	0.00	0.00	501.98	4,838,400.00 *gal*
Application event totals		26.46	7.46	40.92	755.30	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	7.98	0.00	0.00	627.47	6,048,000.00 *gal*
Application event totals		7.98	0.00	0.00	627.47	

5 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.46	14.09	200.71	876.90	864,000.00 *gal*
Ag Well #1	Ground water	6.38	0.00	0.00	501.98	4,838,400.00 *gal*
Application event totals		64.84	14.09	200.71	1,378.88	

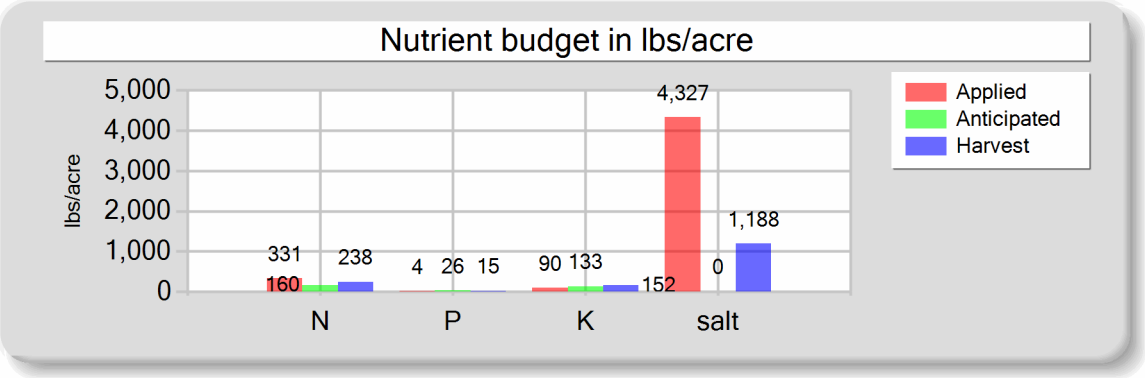
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #1	Ground water	7.98	0.00	0.00	627.47	6,048,000.00 *gal*
Application event totals		7.98	0.00	0.00	627.47	

B. NUTRIENT BUDGET

1 - 11/10/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	307.26	4.39	89.96	3,037.58
Fresh water	16.40	0.00	0.00	1,289.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	330.67	4.39	89.96	4,327.38
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	238.50	15.33	152.40	1,188.18
Nutrient balance	92.17	-10.93	-62.44	3,139.20
Applied to removed ratio	1.39	0.29	0.59	3.64

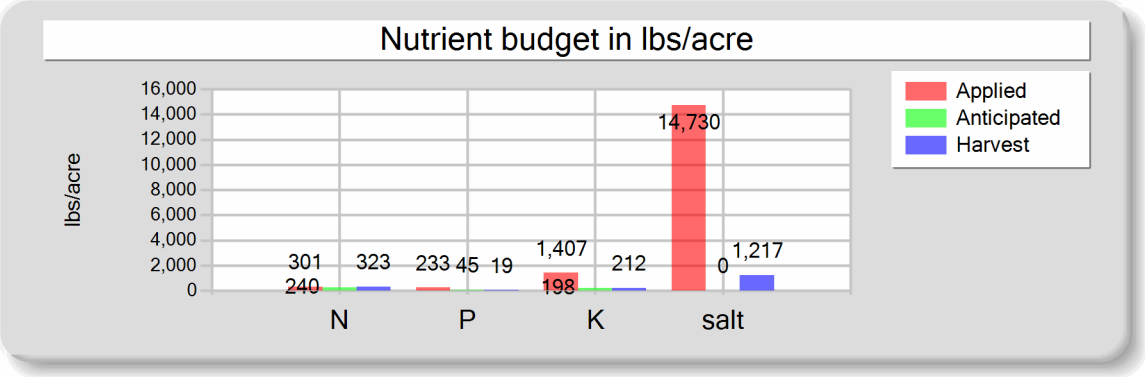
Fresh water applied	
6,048,000.00	*gallons*
222.73	*acre-inches*
12.37	*inches/acre*

Process wastewater applied	
2,520,000.00	*gallons*
92.80	*acre-inches*
5.16	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/17/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	13.90	34.50	34.50	0.00
Dry manure	63.12	155.22	974.34	7,023.10
Process wastewater	143.87	43.67	398.06	1,994.46
Fresh water	72.64	0.00	0.00	5,711.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	300.53	233.39	1,406.90	14,729.54
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	322.56	19.15	211.68	1,216.80
Nutrient balance	-22.03	214.24	1,195.22	13,512.74
Applied to removed ratio	0.93	12.19	6.65	12.11

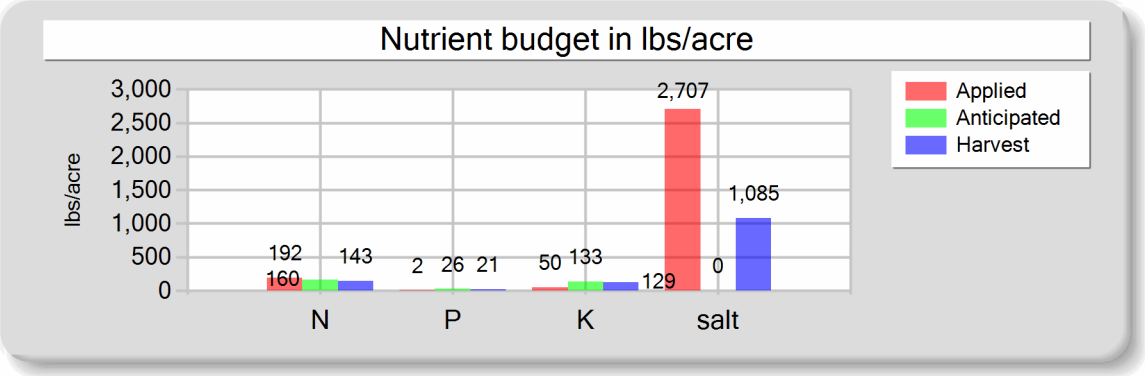
Fresh water applied	
26,784,000.00	*gallons*
986.36	*acre-inches*
54.80	*inches/acre*

Process wastewater applied	
1,980,000.00	*gallons*
72.92	*acre-inches*
4.05	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/10/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	171.76	2.46	50.29	1,698.03
Fresh water	12.84	0.00	0.00	1,009.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.60	2.46	50.29	2,707.44
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	143.05	20.75	128.86	1,085.45
Nutrient balance	48.55	-18.29	-78.57	1,621.99
Applied to removed ratio	1.34	0.12	0.39	2.49

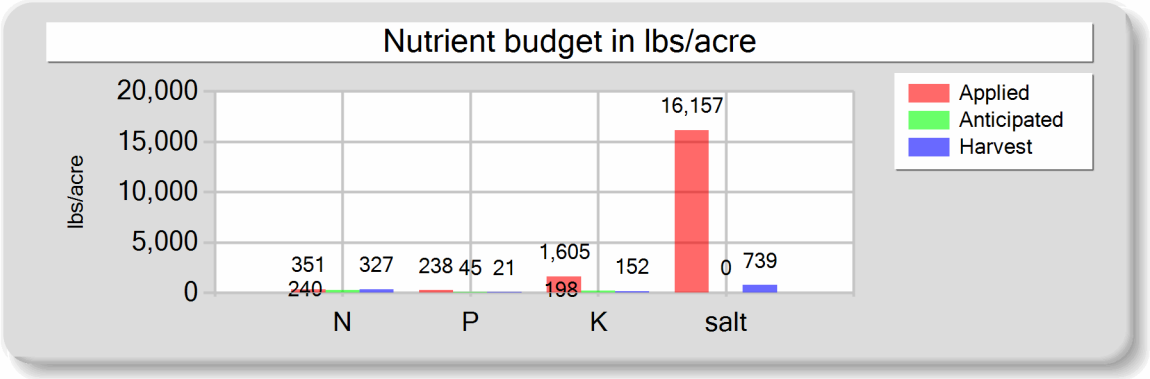
Fresh water applied	
6,048,000.00	*gallons*
222.73	*acre-inches*
9.68	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
2.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/17/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	13.90	34.50	34.50	0.00
Dry manure	63.12	155.22	974.34	7,023.10
Process wastewater	184.99	48.28	595.75	2,707.70
Fresh water	81.73	0.00	0.00	6,426.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	350.74	238.00	1,604.59	16,157.38
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	327.12	21.44	151.73	739.15
Nutrient balance	23.62	216.56	1,452.86	15,418.23
Applied to removed ratio	1.07	11.10	10.58	21.86

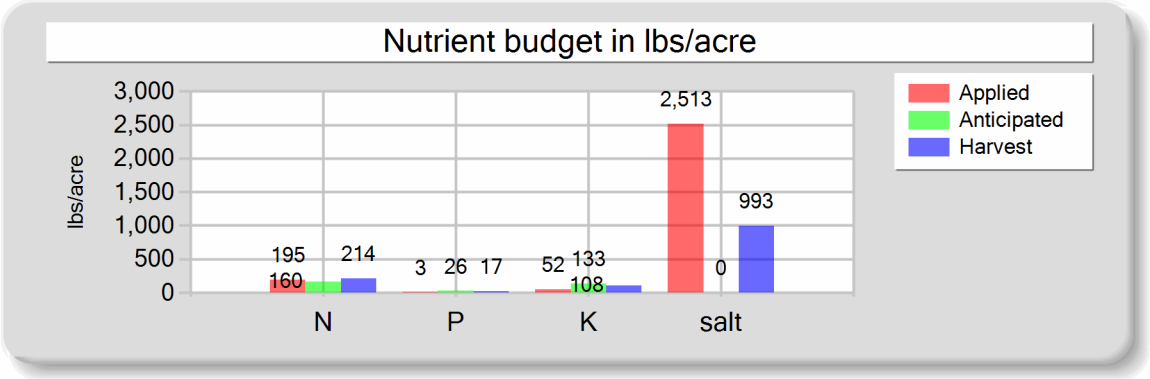
Fresh water applied	
38,505,600.00	*gallons*
1,418.03	*acre-inches*
61.65	*inches/acre*

Process wastewater applied	
2,196,000.00	*gallons*
80.87	*acre-inches*
3.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/12/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	178.41	2.55	52.23	1,763.76
Fresh water	9.52	0.00	0.00	748.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	194.93	2.55	52.23	2,512.67
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	213.59	16.96	107.84	992.56
Nutrient balance	-18.65	-14.41	-55.61	1,520.12
Applied to removed ratio	0.91	0.15	0.48	2.53

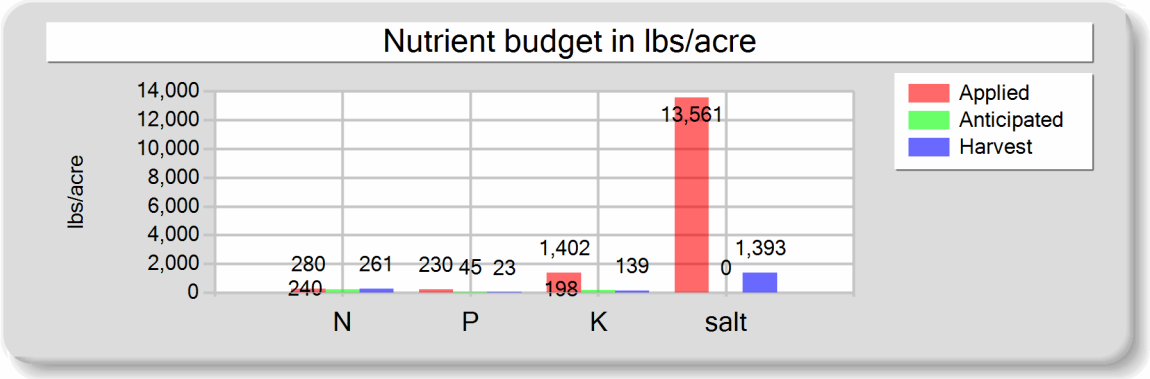
Fresh water applied	
6,048,000.00	*gallons*
222.73	*acre-inches*
7.18	*inches/acre*

Process wastewater applied	
2,520,000.00	*gallons*
92.80	*acre-inches*
2.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/19/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	13.90	34.50	34.50	0.00
Dry manure	63.12	155.22	974.34	7,023.10
Process wastewater	137.30	40.40	393.45	1,926.56
Fresh water	58.64	0.00	0.00	4,611.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	279.96	230.12	1,402.30	13,560.86
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	260.82	23.33	139.32	1,393.20
Nutrient balance	19.14	206.79	1,262.98	12,167.66
Applied to removed ratio	1.07	9.86	10.07	9.73

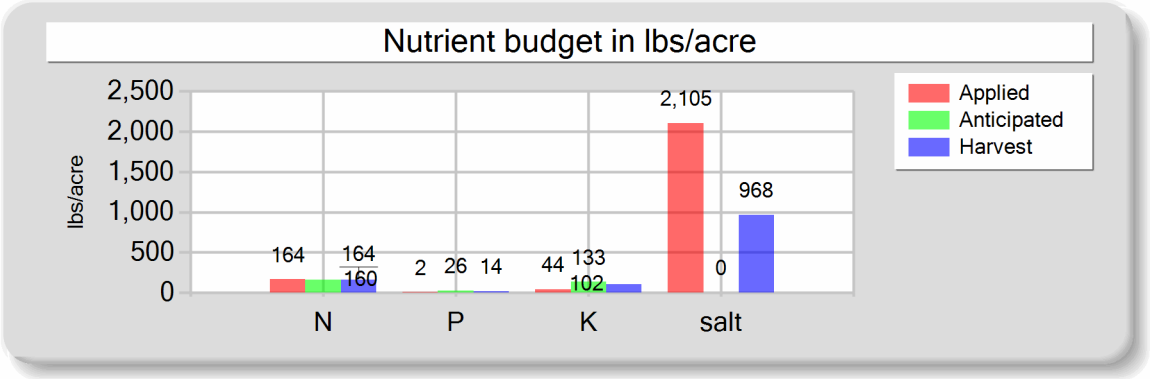
Fresh water applied	
37,238,400.00	*gallons*
1,371.36	*acre-inches*
44.24	*inches/acre*

Process wastewater applied	
3,024,000.00	*gallons*
111.36	*acre-inches*
3.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 11/12/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	149.48	2.14	43.76	1,477.74
Fresh water	7.98	0.00	0.00	627.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	164.46	2.14	43.76	2,105.21
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	163.50	14.10	102.00	967.50
Nutrient balance	0.96	-11.96	-58.24	1,137.71
Applied to removed ratio	1.01	0.15	0.43	2.18

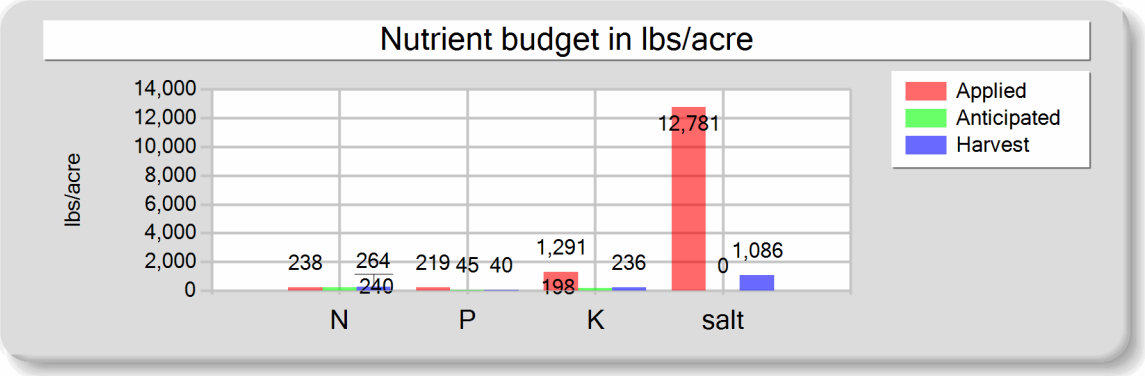
Fresh water applied	
6,048,000.00	*gallons*
222.73	*acre-inches*
6.02	*inches/acre*

Process wastewater applied	
2,520,000.00	*gallons*
92.80	*acre-inches*
2.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 06/18/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	13.90	34.50	34.50	0.00
Dry manure	63.12	155.22	974.34	7,023.10
Process wastewater	98.60	29.02	282.56	1,383.56
Fresh water	55.63	0.00	0.00	4,374.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.25	218.74	1,291.40	12,781.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	263.70	40.08	235.57	1,086.44
Nutrient balance	-25.45	178.65	1,055.83	11,694.59
Applied to removed ratio	0.90	5.46	5.48	11.76

Fresh water applied	
42,163,200.00	*gallons*
1,552.73	*acre-inches*
41.97	*inches/acre*

Process wastewater applied	
2,592,000.00	*gallons*
95.45	*acre-inches*
2.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure 2102606-01

Sample and source description: Manure 2102606-01

Sample date: 10/26/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,550.00	8,730.00	54,800.00	14,000.00	12,000.00	13,000.00	266.00	12,900.00		39.50
DL	0.50	0.50	0.50	5.00	5.00	5.00	1.00	0.50		0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2301426-01

Sample and source description: Lagoon Water 2301426-01

Sample date: 01/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	347.00	193.00			55.40	783.00								4,540.00	4,000
DL	0.60	6.00			0.10	10.00								1.00	1

Lagoon Water 2303487-01

Sample and source description: Lagoon Water 2303487-01

Sample date: 03/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	263.00	18.50			3.76	77.00								4,240.00	2,600
DL	0.60	0.60			0.10	1.00								1.00	1

Lagoon Water 2307290-01

Sample and source description: Lagoon Water 2307290-01

Sample date: 07/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	103.00	75.20			38.30	210.00								2,170.00	1,300
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2310010-01

Sample and source description: Lagoon Water 2310010-01

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	300.00	152.00			72.30	1,030.00								5,220.00	4,500
DL	0.60	0.60			0.10	10.00								1.00	1

C. FRESH WATER ANALYSES

Ag Well #1

2401054-02

Sample description: 2401054-02

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.85										760.00	460
DL	1.40										1.00	1

Ag Well #2

Ag Well #2

2401054-03

Sample description: 2401054-03

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.16										942.00	540
DL	1.40										1.00	1

Domestic Well

2401054-01

Sample description: 2401054-01

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											439.00	
DL											1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 11/10/2022: Oats, silage-soft dough

1 - 11/10/2022: Oats, silage-soft dough

2305223-01

Sample and source description: 2305223-01

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,700.00	1,780.00	17,700.00		13.80
DL	5.00	0.50	0.50		0.10

1 - 06/17/2023: Corn, silage

2311035-01

Sample and source description: 2311035-01

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	1,330.00	14,700.00		8.45
DL	0.50	0.50	0.50		0.10

2 - 11/10/2022: Oats, silage-soft dough

2305223-02

Sample and source description: 2305223-02

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	1,900.00	11,800.00		9.94
DL	0.50	0.50	0.50		0.10

2 - 06/17/2023: Corn, silage

2311035-02

Sample and source description: 2311035-02

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,500.00	1,540.00	10,900.00		5.31
DL	0.50	0.50	0.50		0.10

3 - 11/12/2022: Oats, silage-soft dough

2305223-03

Sample and source description: 2305223-03

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,400.00	1,620.00	10,300.00		9.48
DL	2.50	0.50	0.50		0.10

3 - 06/19/2023: Corn, silage

2311035-03

Sample and source description: 2311035-03

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,100.00	1,440.00	8,600.00		8.60
DL	0.50	0.50	0.50		0.10

5 - 11/12/2022: Oats, silage-soft dough

2305223-04

Sample and source description: 2305223-04

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,800.00	1,880.00	13,600.00		12.90
DL	2.50	0.50	0.50		0.10

5 - 06/18/2023: Corn, silage

2311035-04

Sample and source description: 2311035-04

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	2,280.00	13,400.00		6.18
DL	0.50	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

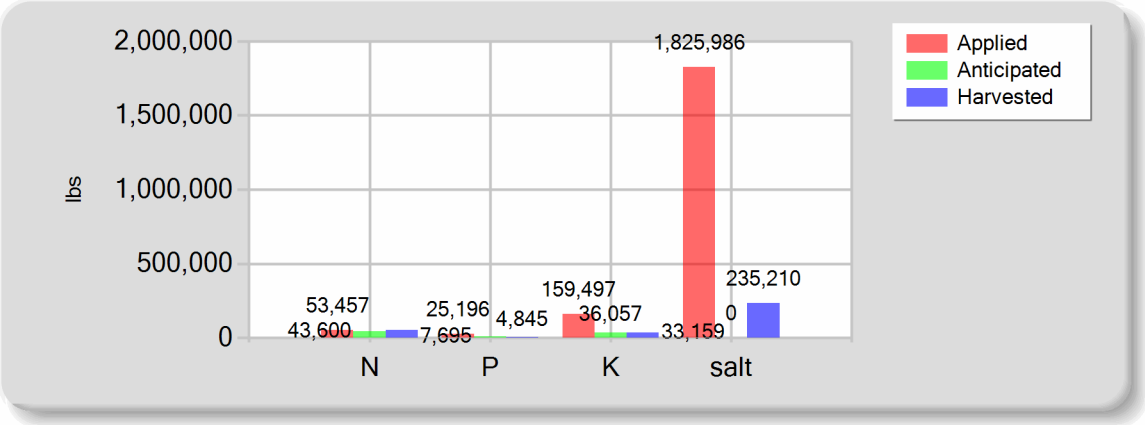
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

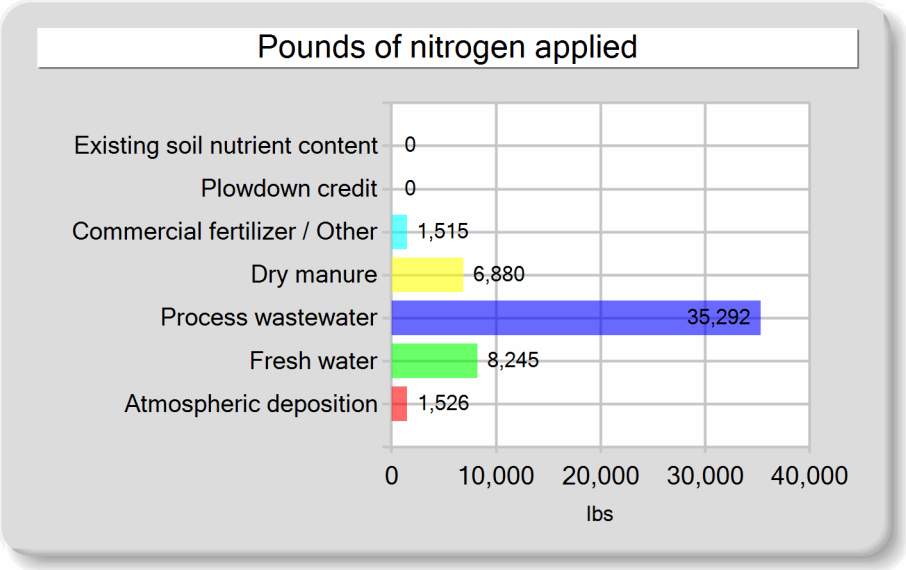
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

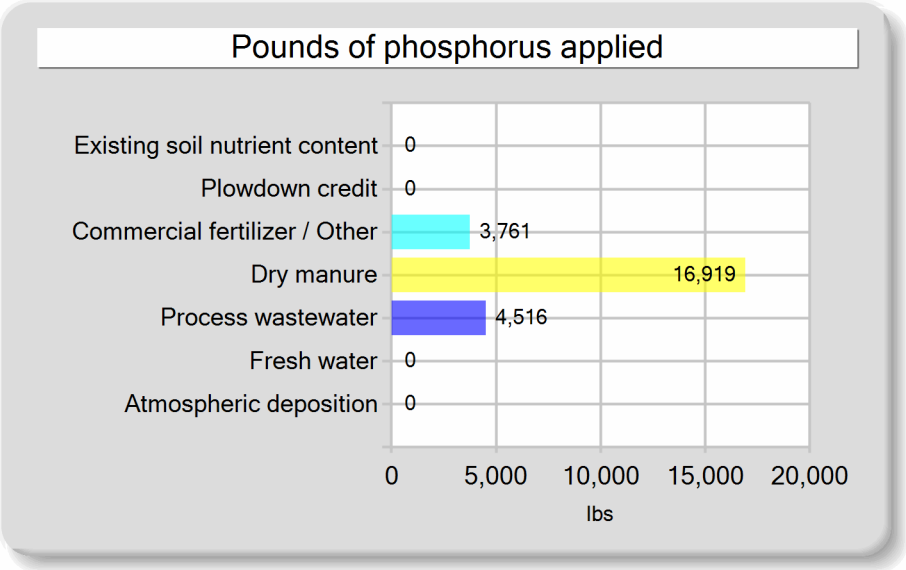
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	1,515.10	3,760.50	3,760.50	0.00
Dry manure	6,879.97	16,918.91	106,203.50	765,517.90
Process wastewater	35,291.84	4,516.27	49,533.25	412,176.24
Fresh water	8,244.58	0.00	0.00	648,291.94
Atmospheric deposition	1,526.00	0.00	0.00	0.00
Total nutrients applied	53,457.49	25,195.69	159,497.25	1,825,986.08
Anticipated crop nutrient removal	43,600.00	7,695.40	36,057.20	0.00
Actual crop nutrient removal	51,426.03	4,844.57	33,158.97	235,209.80
Nutrient balance	2,031.46	20,351.12	126,338.27	1,590,776.28
Applied to removed ratio	1.04	5.20	4.81	7.76

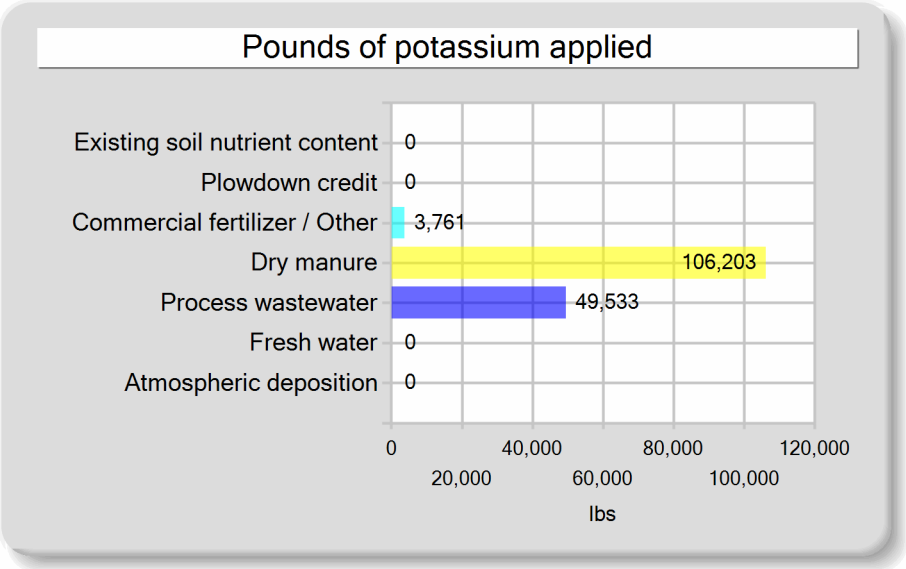
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

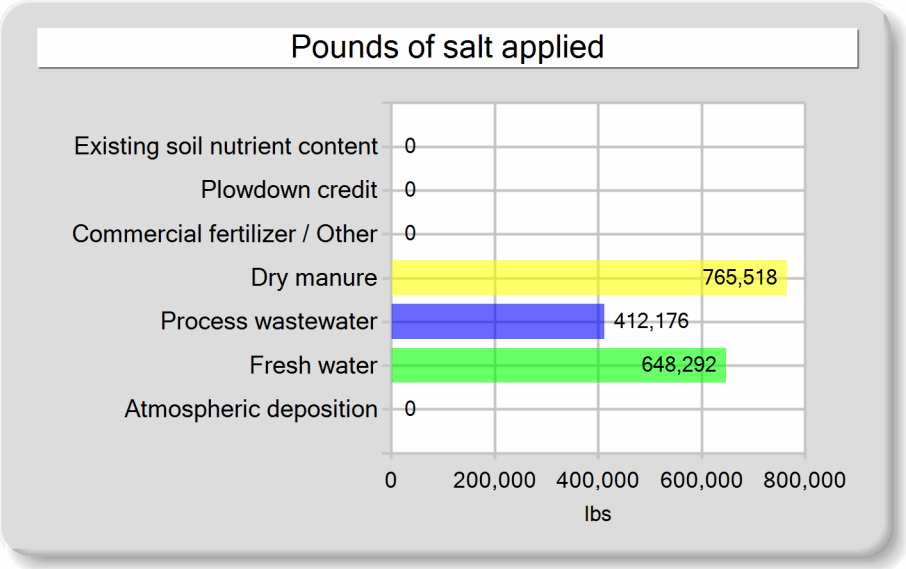


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

X

SIGNATURE OF OWNER OF FACILITY

Tony & Filemina Nunes/E&L Land Group

PRINT OR TYPE NAME

Edward Pimentel

DATE

X

SIGNATURE OF OPERATOR OF FACILITY

Eddie Pimentel

PRINT OR TYPE NAME

Edward Pimentel

DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	N/A
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	